NLP 201 - Natural Language Processing
Assignment 1

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1 Question 1

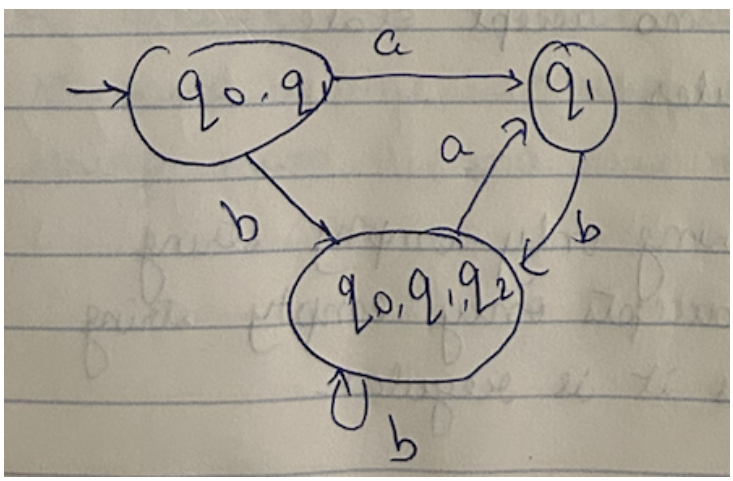

Figure 1: Question 1

2 Question 2

DFA depiction of both the parts.

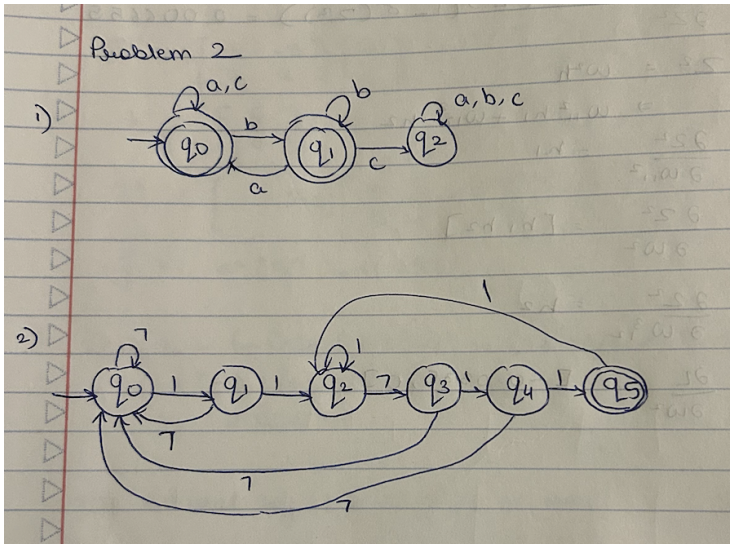

Figure 2: Question 2

3 Question 3

Non Deterministic FSA, below are the images.

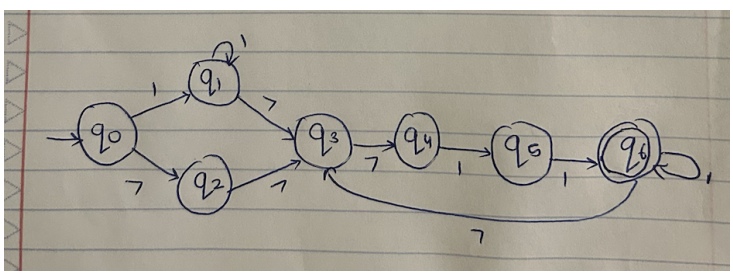

Figure 3: Question 3

4 Question 4

1. Set of strings over {1,7} ending 711: $[1|7] * 117$
2. Set of strings over {a,b,c} containing 'ab' or 'bc': $[a|b|c] * (ab|bc)[a|b|c]*$

3. Set of strings over {a,b,c} either starting with 'a' and containing the sub string 'ab' or starting with 'b' and containing the sub string 'bc': $[a * (a|b|c) * ab(a|b|c) * |b * (a|b|c) * bc(a|b|c)*]$

5 Question 5

- $(\alpha \wedge \beta)$ is a GRE denoting $A \cap B$;

  The language that would have strings from A and B. Thus if A and B are regular, their intersection would be regular too.

- $\neg\alpha$ is a GRE denoting $\overline{A}$. The language that contains all strings not in A. Here, because we can construct a finite automaton for the complement of a regular language and that complement is still regular, we conclude that the language denoted by $\overline{A}$ is regular.

6 Question 6

1. Let $L_1 \subset L_2$. If L_1 is not regular, then L_2 must also be not regular : False

   $\Sigma = \{a, b\}$
   $L_1 = \{a^n b^n\}$
   $L_2 = \Sigma^*$

   Here, L_2 is all possible strings over Σ, and $L_1 \subset L_2$.

2. $L = L_1 \cap L_2$. If L_1 and L are regular languages, then L_2 must also be a regular language : False

   L = regular {ab}
   $L_1 = regular\{ab\}$
   $L_2 = \{a^n b^n\}$

   Here, L_2 is not a regular language.

3. $L = L_1 \cup L_2$. If L_1 and L are regular languages, then L_2 must also be a regular language : False

   $\Sigma = \{a, b\}$
   $\Sigma^* = Allstringsover\Sigma = aregularlanguage$
   $L = L_1 \cup L_2 = \Sigma^*$

Here, L_2 might not be regular. For example, if $L_2 = \{a^n b^n\}$, then it is not a regular language.

4. $L = \bigcap_{i=1}^{\infty} L_i$. If all of the L_i are regular languages, then L is also a regular language: False

   $\Sigma = \{a\}$
   $\frac{\Sigma^*}{a^i} = Allstringsin\Sigma^* dividedbya^i$
   $\Sigma^* = Allstringsover\Sigma, whichisaregularlanguage$
   $L = L_1 \cup L_2 = \Sigma^*$

   Here, $L = \bigcap_{i=1}^{\infty} L_i$ removes strings like a^i. Assuming we define L to avoid strings like $a^n b^n$, it might not be regular.

5. Let α, β, and γ be regular expressions.
   If $L(\beta \mid \alpha\gamma) \subseteq L(\gamma)$, then $L(\alpha^*\beta) \subseteq L(\gamma)$: True

$$\Sigma = \{a, b\};\ \alpha = a;\ \beta = b;\ \gamma = ab$$

$$L(\beta|\alpha\gamma) = L(b|aab)whichhasasubsetL(ab)$$

$$ForL(\alpha^*\beta) = L(a^*b)wehaveL(\gamma) = abthatisasubset$$